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Architecting Human-Like Operational Intelligence for Drilling Interventions Through Generative Artificial Intelligence

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Abstract

In today's complex oil and gas drilling landscape, timely access to past operational knowledge is critical for safe and efficient decision-making. This work introduces a system that combines real-time intervention capture with generative artificial intelligence to bridge the gap between historical data and immediate field needs. Operational interventions are automatically logged and communicated through a custom platform, ensuring that key parameters are preserved and integrated into a unified, traceable backend database.

A generative artificial intelligence (AI) layer, built around the Mistral 7B model, provides intelligent access to this database. Instead of requiring users to sift through static dashboards or reports, the system retrieves and interprets relevant historical records based on context. The model processes operational prompts to generate tailored summaries, classifications, or recommendations, offering responses that mirror the reasoning of experienced professionals. This contextual interaction improves situational awareness while minimizing ambiguity and manual workload.

The approach marks a shift in how drilling operations interact with data, from passive retrieval to proactive, AI-driven support. By transforming operational records into dynamic decision aids, the system enhances both speed and accuracy in field responses. More than a functional upgrade, this platform sets the stage for cognitive operations where machines actively contribute to strategic execution through human-like understanding and interaction.

Introduction

Drilling operations produce an overwhelming amount of digital information, primarily sourced from a wide array of surface and downhole sensors installed across the rig site. These data streams offer a largely untapped potential to improve operational efficiency. In recent years, the industry has increasingly focused on integrating data from multiple channels, including surface drilling equipment, mud logging units, and downhole measurement tools, in an effort to drive real-time optimization and informed decision-making.

In current oil and gas drilling, operations are becoming increasingly complex due to factors like deeper wells and unconventional reservoirs. Having access to historical data and operational practices is key to optimize well construction processes and drill wells safely. Drilling operations produce an overwhelming amount of digital information, primarily sourced from a wide array of surface and downhole sensors

installed across the rig site. These data streams offer a largely untapped potential to improve operational efficiency. In recent years, the industry has increasingly focused on integrating data from multiple channels, including surface drilling equipment, mud logging units, and downhole measurement tools, in an effort to drive real-time optimization and informed decision-making. Yet, many operations still face gaps in data management practices, hindering their ability to fully leverage the potential of digital drilling data ([Isbell et al. 2022](#)). This ambitious integration effort introduces significant technical and organizational challenges. The sheer volume, velocity, and heterogeneity of the datasets can strain existing systems and workflows. When data is scattered across different platforms and formats, it becomes difficult to retrieve and analyze quickly. This fragmentation leads to delays in decision making and increases the risk of oversight.

Over the past decade, the oil and gas (O&G) industry has faced intense pressure to enhance efficiency, particularly in drilling operations. Drilling engineers are increasingly tasked with leveraging vast amounts of high-frequency sensor and reporting data to optimize designs and processes. While efficient data access is crucial for upstream projects, data operations remain a skill gap within the drilling engineering community, limiting their ability to fully harness the potential of these datasets ([Amour et al. 2024](#)). Extracting insights from large datasets is critical for energy companies to optimize operations across various business segments. Intelligent technologies enable efficient management and analysis of complex data, driving informed decision-making. ([Singh et al. 2023](#))

To enhance drilling performance, companies have increasingly relied on Real-Time Operation Centers (RTOCs), where experts monitor live drilling data and provide immediate guidance. However, as drilling activity expands, the sheer volume of data surpasses human analytical capacity, limiting the effectiveness of RTOCs and creating a need for AI and ML-based solutions ([Al Gharbi et al. 2021](#)). The adoption of Artificial Intelligence in drilling optimization is advancing rapidly, with applications increasingly focused on automating processes, reducing costs, mitigating risks, and enhancing overall operational performance ([Creegan and Roberts 2021](#)). Industry 3.0 centered on process automation, whereas Industry 4.0, powered by AI, is shifting the focus toward automating decision-making ([Payrazyan et al. 2025](#)). Recent applications show that Generative AI can deliver up to 99% faster processing of drilling data, demonstrating its potential to revolutionize not only mud reporting but also large-scale operational workflows across the drilling industry ([Wang et al. 2025](#)). Integrating Generative AI into drilling requires a domain-driven, agent-based methodology tailored to industry-specific terminology, data sources, and operational complexity ([Ponomareva et al. 2024](#)).

In cases, when historical real time data is fragmented and not readily available, Operations team can face the following:

- Repeat errors due to lack of visibility into past operational issues and practices
- Miss opportunities to improve efficiency and reduce Invisible Lost Time (ILT)
- Spend extra time searching for or validating the information.

To overcome such scenarios, a system was designed and implemented on the basis of real time data which was received from Rig site to central hub. After data streaming, real time interventions were captured based on the well operation and abnormality classification. Each intervention is documented with relevant details (e.g. event time, event type, section, and other contextual information), creating a structured and searchable database. This database becomes a learning tool for future wells, enabling proactive planning. Consistent formatting ensures that users can follow the lifecycle of each intervention from detection to resolution, supporting future reviews, and continuous improvement. Lastly, generative AI was leveraged in this work to interpret and contextualize the captured operational data, enabling the automated generation of summaries, recommendations, and even predictive insights derived from historical patterns. Unlike traditional rule-based systems, generative AI models are capable of understanding complex relationships in

unstructured and semi-structured data, making them especially valuable in domains with high variability and nuance, such as drilling operations. These models can synthesize large volumes of information, detect subtle anomalies, and communicate findings in natural language, bridging the gap between raw data and human decision-making. Nonetheless, it is important to recognize certain limitations. Large Language Models (LLMs) do not possess genuine human-like understanding; instead, they approximate comprehension by capturing statistical patterns in data. This often results in hallucinations, where models generate outputs that appear plausible but are factually incorrect or unsupported by the underlying records. Moreover, their reliability is strongly influenced by how information is structured and presented. However, generative AI enhances scalability by reducing manual interpretation efforts, accelerates insight generation through real-time responses, and supports continuous learning by adapting to new data over time. As a result, it introduces a paradigm shift in how engineering teams interact with operational data; moving from static reporting toward dynamic, AI-driven understanding.

Methodology

Our methodology is structured into two primary components. The first focuses on the foundational aspects of real-time intervention management, capturing operational data at the point of occurrence, ensuring consistent ingestion, and enabling seamless communication with stakeholders. The second leverages generative AI to transform historical records into dynamic, context-aware decision support outputs. Together, these components create an intelligent workflow that bridges real-time operations with expert-level analysis.

High-Level Methodology Overview: From Field Events to Intelligent Decisions

This methodology represents a modern, end-to-end approach to intervention data management and insight generation. It begins with the real-time generation and capture of operational interventions through a custom-built platform, designed to log critical parameters from field environment. These structured records are simultaneously routed to relevant stakeholders through automated notification systems, creating a synchronized flow of operational awareness. A centralized backend ingestion layer ensures data uniformity, integrity, and traceability, regardless of the field conditions or environments involved.

Building on this data foundation, a generative AI layer is introduced to unlock deeper insight. Using a context-aware retrieval engine, the system identifies top-matching historical interventions, which are then processed by the Mistral 7B large language model—fine-tuned for instruction-following tasks. The model interprets operational context and produces expert-level outputs including narrative summaries, pattern classifications, and actionable recommendations. This architecture turns static data into an intelligent, adaptive decision-support ecosystem.

Core System Element One: Intelligent Infrastructure for Real-Time Intervention Management

Collecting and Transferring Real-Time Data. An integrated data acquisition system enhances the drilling process, starting with real-time sensors located in the Bottom Hole Assembly (BHA). These sensors continuously capture essential downhole parameters such as pressure, temperature, vibration levels, and fluid flow rates as the BHA progresses through the wellbore. Complementing these are surface-based instruments that record additional key indicators like weight on bit (WOB), rotational speed (RPM), and surface flow rates, collectively offering a detailed snapshot of the drilling process.

All collected sensor data is routed to a localized data collection unit, often referred to as the "black box." This unit is engineered to process substantial volumes of high-frequency data sampled at a minimum frequency of 0.2 Hz, handling its encryption and preparing it for secure transmission via high-speed communication channels to a centralized operations center. The low-latency transmission enables near-instantaneous access to real-time drilling data.

Upon arrival at the central hub, the data is visualized using advanced drilling analytics platforms. Specialists monitor these data streams continuously, watching for patterns or irregularities. If deviations from normal operating behavior are detected, the system issues immediate alerts, allowing for swift corrective actions. Featuring intelligent anomaly detection and real-time data processing, these platforms support rapid, data-driven decision making, compressing a 10 minute email driven task into just 2 minutes on average. This represents an 80 percent efficiency gain and a five times speed improvement. Unlike manual interventions through email, which are prone to errors such as a crew monitoring multiple rigs mistakenly sending actions to the wrong channel, the platform ensures efficiency and accuracy while minimizing the risk of costly delays and operational mistakes.

Structured Interface for Intervention Logging. To enhance the efficiency and responsiveness of the intervention process, the methodology presents a tailored platform, depicted in Fig.1 below, that seamlessly combines real-time intervention data with a backend database. This platform simplifies data handling, enabling faster and more accurate interventions. With an intuitive interface, monitoring personnel can quickly log essential details like time, depth, operation type, and intervention level.

The figure displays two screenshots of the 'Intervention Application' user interface. The top screenshot shows a form with the following fields: Client (Select Client), Submitted by (Select Name), Submitted at (Hours, Min), Intervention Level (Select Level), Event Type (Select Event Type), Operations (Select Operations), Section (Select Section), Field Name (Select Field Name), Well Name (Select Well Name), Rig Name (Select Rig Name), Communicated to Rig (Yes), Event Start Time (Hours, Min), Event End Time (Hours, Min), Data Provider (Select Data Provider), Depth (Enter Depth, Unit), and Comments. The bottom screenshot shows the same form with additional fields for Data Provider, Depth, and Comments, and buttons for Select Image, Upload Image, Send Email, and Save.

Figure 1—Intervention Application User Interface

Automated Communication Pipeline for Stakeholder Synchronization. The system enhances operational efficiency by automating communication workflows, specifically through the generation and distribution of real-time intervention through email. The system sends detailed intervention notifications to designated stakeholders, including drilling engineers, supervisors, and operations managers, ensuring that all relevant parties are immediately informed of any critical developments. This timely dissemination of information enables rapid response and intervention, minimizing delays and potential risks. By facilitating seamless, real-time coordination across teams through swift email communication, the system supports informed decision-making, accelerates problem resolution, and strengthens overall operational oversight.

Centralized Storage with Contextual Metadata Architecture. The intervention data is consolidated in a centralized repository, designed with a layered metadata architecture that captures contextual information such as source, timestamp, drilling phase, etc. This structure not only enhances data discoverability but also supports advanced querying and contextual analytics, improving operational insights and knowledge retention.

Core System Element Two: Generative AI Layer for Contextual Insight and Operational Intelligence

To enable context-aware analysis and generate useful insights from historical intervention data, we built a generative AI pipeline that combines semantic indexing, efficient retrieval, structured prompting, and inference using a pre-trained instruction-tuned language model.

Before describing each component in detail, [Fig.2](#) below provides an overview of the Retrieval-Augmented Generation (RAG) architecture implemented in this study. This architecture grounds the output of a large language model in historical intervention data by pairing retrieval (semantic search over a vector database) with generation (natural-language responses). This reduces reliance on the model's parametric memory and ensures that generated answers are anchored in domain-specific evidence.

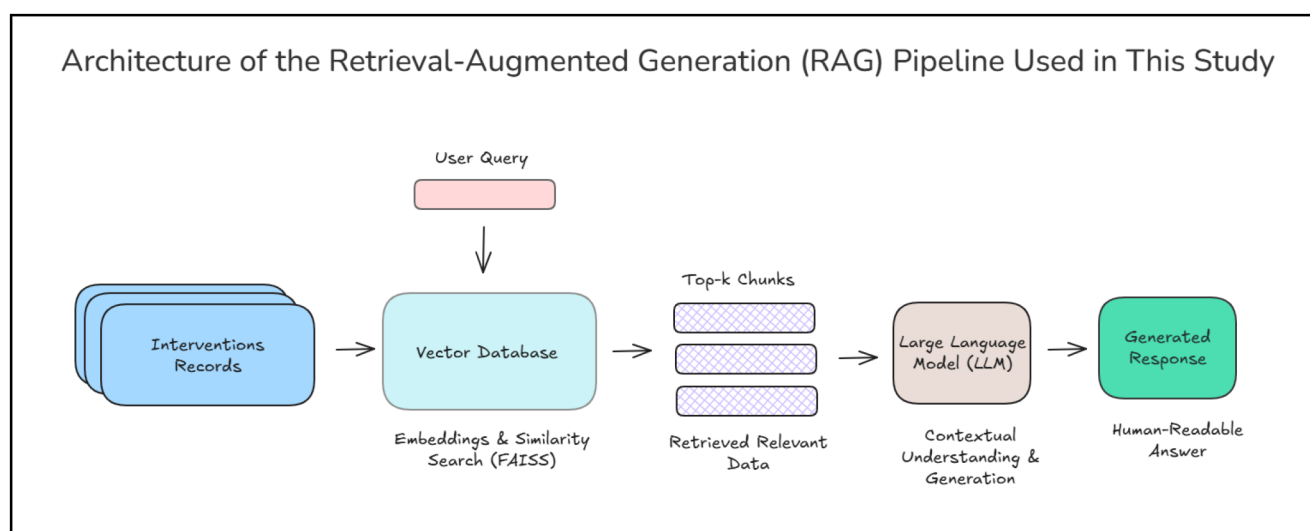


Figure 2—Architecture of the Retrieval-Augmented Generation (RAG) pipeline used in this study.

As shown in the figure, the system operates in five sequential stages:

1. **Embedding & Vector Database Construction:** Historical intervention records are embedded using a sentence-transformer model ([Reimers et al., 2019](#)) and stored in a vector database for efficient retrieval.
2. **Query Embedding & Top-K Retrieval:** A user question is embedded in the same vector space, and FAISS (Facebook AI Similarity Search; [Johnson et al., 2021](#)), an open-source similarity search library

originally developed by Facebook AI Research and now maintained by Meta AI, returns the top-k most semantically similar records.

3. **Prompt Construction:** The retrieved records are compiled with the user's question into a structured prompt that delineates context from inquiry.
4. **Language Model Inference:** The prompt is processed by an instruction-tuned Mistral 7B model (GGUF format) (Jiang et al. 2023), which generates human-like, context-grounded interpretations.
5. **Response Generation:** The final output is returned to the user as a natural-language answer, supported by the underlying retrieved records.

This end-to-end flow provides the foundation for the detailed components described below.

Semantic Indexing and Embedding of Historical Interventions. To enable context-aware analysis of historical interventions, the dataset, originally in tabular CSV format, is preprocessed into line-by-line textual records. Each row is transformed into a unified string by concatenating its column values, preserving both structure and content. These text lines are then semantically embedded using the all-MiniLM-L6-v2 model from the Sentence Transformers library as shown in Fig.6 in the appendix. This step translates human-readable intervention records into fixed-size high-dimensional vectors that preserve meaning, allowing the system to reason about the data in a concept-driven rather than keyword-driven manner. The resulting embeddings form the basis of all downstream retrieval and reasoning processes.

Relevance-Optimized Retrieval Engine for Context Matching. Once the embeddings are generated, they are indexed using Facebook AI Similarity Search (FAISS), an open-source library developed by Facebook AI Research designed for efficient similarity search and clustering of high-dimensional vectors (Fig.7 in the appendix). This creates an efficient retrieval layer that can return the most contextually relevant entries in response to user queries. When a user inputs a question, it is also embedded into the same vector space, and the FAISS engine identifies the top-k nearest neighbor vectors. These represent the most semantically aligned records from the dataset. This relevance-optimized search mechanism ensures that only the most pertinent data is considered during inference, improving both precision and interpretability of the generated output.

Operational Prompt Engineering for Contextual AI Interpretation. The retrieved entries are compiled into a structured prompt (Fig.8 in the appendix), designed to guide the language model's response. This prompt includes both the retrieved context and the user's question, formatted to clearly delineate informational input from the inquiry. This approach to prompt engineering minimizes ambiguity, allowing the model to focus on generating responses grounded in the supplied evidence. The prompt format was developed to suit instruction-tuned models and maintain consistent performance across different question types, including descriptive, analytical, and summarization tasks.

Language Model Inference for Human-Like Interpretation. The structured prompt is processed by an instruction-tuned Mistral 7B (Jiang et al. 2023) language model, which interprets the retrieved intervention records in light of the user's question. Operating in GPT-Generated Unified Format (GGUF) format for efficient inference, the model generates context-aware, natural-language responses that reflect expert-like reasoning. Depending on the query, the output may include summaries, classifications, explanations, or descriptive narratives.

Insight Generation Module for Summaries and Recommendations. The final layer of the system uses semantic retrieval to identify the most relevant historical intervention records based on the user's query. These top-matching entries, retrieved via FAISS and sentence embeddings, are directly passed to the language model without additional context structuring or prompt engineering. The model interprets these raw text segments as-is and generates a response aligned with the original question. Depending on how the question is framed, the model's output may range from simple lookups to high-level summaries, but its

quality is constrained by the implicit context embedded within the retrieved text rather than explicit prompt guidance. This lightweight design enables rapid querying but leaves room for refinement through more structured context injection and prompt scaffolding.

As shown in [Fig.3](#), the instruction-tuned Mistral 7B language model is prompted with a domain-specific question: "Which drilling interventions are most significant and why?"

```
print(ask_question("which drilling interventions are most significant and why?"))
```

The most significant drilling interventions are those with a "High" intervention level, as they indicate a more serious issue that requires immediate attention. Additionally, the interventions that have the highest SPP (Surface Pressure) increase are also important, as they indicate that the well is experiencing a significant amount of pressure during drilling. Overall, it's important to closely monitor and address any significant drilling interventions in order to ensure the safety of the rig and surrounding environment.

Figure 3—User Query Example

The model interprets the relevant intervention data and responds with a human-like explanation that highlights both the severity indicator ("High" intervention level) and specific technical metrics (Surface Pressure increase). This demonstrates the model's ability to synthesize structured operational data into expert-like narratives, making it suitable for analytical and decision-support use cases.

Findings and Discussion

Transformation of Intervention Management

The deployment of a unified intervention platform has significantly streamlined the documentation and communication of drilling events. Real-time data acquisition from both downhole and surface sensors is seamlessly integrated into a centralized repository, ensuring high fidelity and traceability. The intuitive user interface allows for rapid logging of critical parameters such as depth, operation type, and intervention level, while automated notifications ensure that stakeholders are promptly informed. This has led to:

- Improved operational awareness through synchronized data flows.
- Faster decision-making enabled by real-time alerts and structured logging.
- Enhanced traceability and analytics via a metadata-rich backend database.

[Fig.4](#) illustrates the detailed breakdown of operational events that is now possible due to categorical classification. In addition, it allows for the creation and maintenance of databases for continuous operational improvements based on targeted actions on common event types.

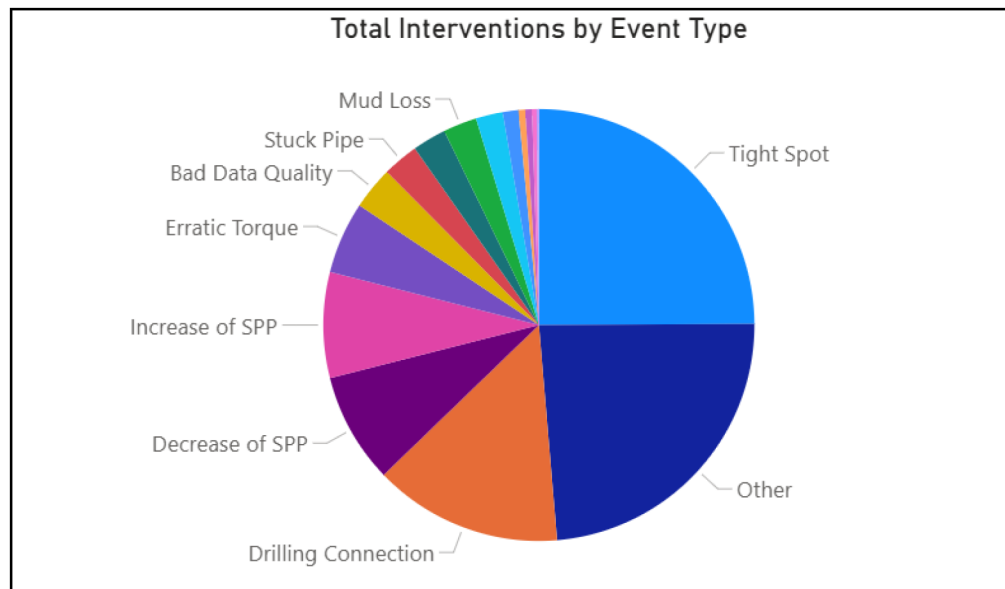


Figure 4—Interventions classified into drilling event types

Generative AI as a Decision Support Catalyst

The integration of generative AI, specifically the Mistral 7B (Jiang et al. 2023) instruction-tuned model, has elevated the system from a passive data repository to an active decision-support tool. By embedding historical intervention records and enabling semantic search through FAISS, the system retrieves contextually relevant data and generates human-like insights. Key outcomes include:

- Context-aware recommendations that mirror expert reasoning.
- Automated summarization and classification of interventions.
- Reduction in manual workload and ambiguity in operational analysis.

This shift from static dashboards to dynamic, AI-driven interaction has improved situational awareness and operational agility.

Enhanced Workflow Optimization

Generative AI has demonstrated its capability to analyze patterns, detect anomalies, and prioritize interventions based on criticality. This has led to:

- Proactive risk mitigation through predictive modeling.
- Optimized intervention strategies tailored to specific operational contexts.
- Scalable cross-well comparisons and retrospective analysis.

The system's ability to simulate scenarios and provide actionable guidance positions as a strategic asset in drilling operations.

Fig.5 depicts an example of the importance of the various events that are detected by the system in one of the well. About 25% of the events were related to tight spot detection for a well that had wellbore stability issues, this type of detection early-on is extremely important to prevent large NPTs.

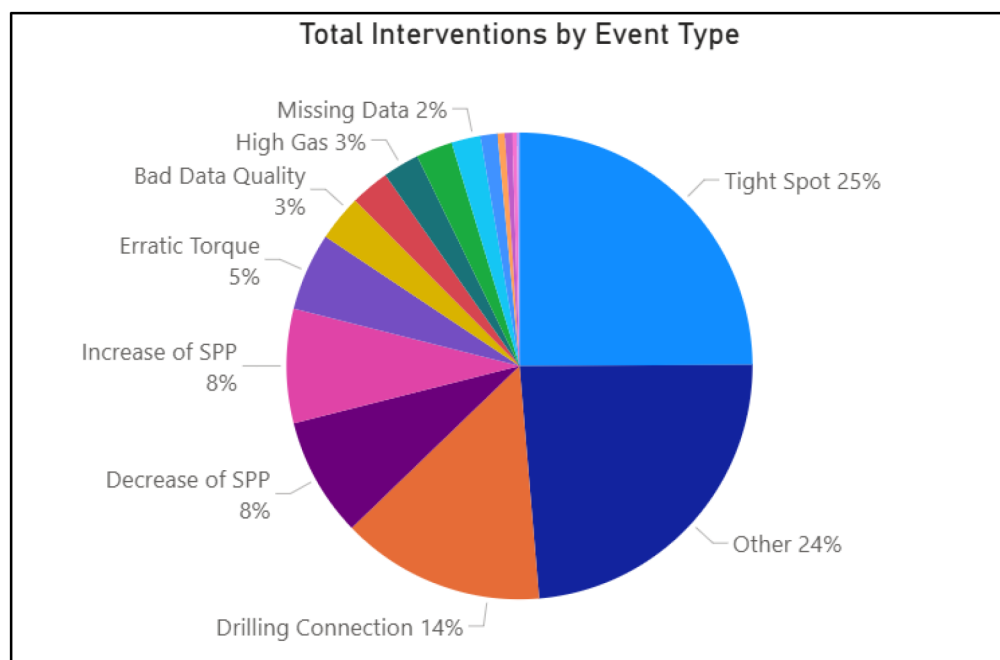


Figure 5—Interventions classified into drilling event types with percentages

Future Potential and Scalability

The architecture lays the groundwork for future enhancements, including:

- Conversational interfaces for interactive querying and guidance.
- Advanced natural language generation for nuanced recommendations.
- Integration with virtual assistants to further reduce response times and improve user engagement.

As generative AI continues to evolve, its role in operational intelligence will expand, driving efficiency, safety, and strategic execution across the oil and gas sector.

Conclusion

This work marks a pivotal shift in how intervention management is approached within drilling operations — moving from fragmented data interpretation to a cohesive, intelligence-driven system. By embedding semantic search and generative AI into the core architecture, we have created more than a tool: we have laid the foundation for a real-time, context-aware decision-support ecosystem.

Technically, the system's integration of FAISS-based similarity search with a instruction-tuned Mistral 7B model showcases how scalable open-source components can deliver high-performance intelligence.

More importantly, the generative AI component does not merely automate responses; it mimics expert cognition, offering nuanced interpretation of historical records, enabling contextually relevant recommendations, and reducing the ambiguity inherent in human decision-making. What once required manual review and subject-matter intervention is now surfaced instantly in a traceable, structured, and explainable method.

Looking forward, this architecture is not the end state, but a launchpad. As generative models become more multimodal, conversational, and process-aware, the opportunity exists to embed AI even more deeply into drilling workflows. Integration with virtual agents, scenario simulators, and closed-loop automation systems could reduce non-productive time further and bring drilling performance closer to autonomous optimization.

To achieve this vision, however, two parallel tracks must advance:

1. The growth and diversification of intervention data:

Current performance is closely tied to the size and representativeness of the historical intervention dataset. Expanding the volume of structured, metadata-rich intervention logs across different well types, geologies, and operational environments will allow the system to generalize better, identify more subtle patterns, and provide higher-confidence recommendations. The addition of labeled intervention outcomes will further unlock supervised fine-tuning and evaluation of AI-generated guidance.

2. The evolution of generative models:

As foundation models continue to advance in reasoning ability, domain adaptability, and memory retention, future iterations could support continual learning, agentic planning, and temporal reasoning, allowing not just retrieval-based answers but simulation of operational outcomes, dynamic prioritization of risks, and alignment with drilling goals in real-time.

In essence, this research shows that the convergence of structured operational data and generative AI is not only feasible, but foundational to the next generation of intelligent drilling systems. It offers a path toward smarter interventions, faster decisions, and more resilient operations, where AI is not just a support layer, but a strategic partner in execution.

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Nomenclature

ILT: Invisible Lost Time
 BHA: Bottom Hole Assembly
 AI: Artificial Intelligence
 WOB: Weight on Bit

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Appendix

```
df = pd.read_csv("/Interventions.csv")
text_lines = df.astype(str).apply(lambda row: " | ".join(row.values), axis=1).tolist()
```

```
embedder = SentenceTransformer('all-MiniLM-L6-v2')
embeddings = embedder.encode(text_lines, convert_to_numpy=True)
```

Figure 6—Indexing and embedding of intervention data

```
dimension = embeddings.shape[1]
index = faiss.IndexFlatL2(dimension)
index.add(embeddings)
```

Figure 7—FAISS index

```
def ask_question(question, top_k=5):
    question_emb = embedder.encode([question], convert_to_numpy=True)
    _, indices = index.search(question_emb, top_k)
    context = "\n".join([text_lines[i] for i in indices[0]])

    prompt = f""Use the following context to answer the question:
{context}

Question: {question}
Answer: ""

    response = llm(prompt, max_tokens=500)
    return textwrap.fill(response['choices'][0]['text'].strip(), width=100)
```

Figure 8—Prompt Code for Contextual AI Interpretation